

Controlling Devices Through WebIOPi Interface



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In the evening, with the day's work over, your office is locked and you are about to leave for home. Suddenly, you notice that some of the lights in your office are still on, which you can see from the factory gate. You open the browser in your laptop, do authentication, give a command, and all the lights in your office are switched off. You return home peacefully.

Take another case. You have gone somewhere and realise that you will not be returning to your office for quite some time. So you go to a nearby computer, open the browser, check the temperature in your office, press some keys on the computer and the AC in your office gets switched off. You can do all this with the help of Raspberry Pi, using WebIOPi software.

WebIOPi

WebIOPi is browser-based software through which the above-mentioned tasks can be done easily yet economically. The latest WebIOPi version 0.0.6 is available at:

<http://code.google.com/p/webiopi/downloads/detail?name=WebIOPi-0.0.6.0.tar.gz&can=2&q=>

Download it with 'wget' command or directly from the browser. Extract the tarball file in a directory using the command (also refer Fig. 1):

```
$ tar xzvf WebIOPi-0.0.6.0.tar.gz
```

Now use 'cd' command to go to the extracted directory and run the following command (also refer Fig. 2):

```
$ sudo ./setup.sh
```

WebIOPi will be installed in your Raspberry Pi. To start WebIOPi, run the following command (also refer Fig. 3):

```
$ sudo /etc/init.d/webiopi start
```

This will not make WebIOPi start

automatically at every boot. To start it automatically at the boot itself, give the command (also refer Fig. 4):

```
$ sudo update-rc.d webiopi defaults
```

Accessing Raspberry Pi's GPIOs

The WebIOPi is essentially an Apache webserver framework with facilities



Fig. 1: Extracting the tarball file



Fig. 2: Installation of WebIOPi



Fig. 3: Starting WebIOPi



Fig. 4: Starting WebIOPi at reboot



Fig. 5: Login to WebIOPi



Fig. 6: WebIOPi main menu

of Raspberry Pi GPIOs. If you already have Apache installed in your Raspberry Pi, just go ahead and WebIOPi will manage itself accordingly. If you do not have Apache, you can install it with the command:

```
$ sudo apt-get install apache2
```

Once WebIOPi is installed, go to the browser window of your computer and open the link:

<http://your-raspi-ip-address:8000/webiopi>

It will ask for a login and password, which you can change later. But for the time being type *webiopi* and *raspberrypi* as login and password, respectively, as shown in Fig. 5.

You will see the WebIOPi main menu, as shown in Fig. 6.

Click on the GPIO Header link and you will see the GPIO header pins along with the pin functions, as shown in Fig. 7. The 'IN' means the corresponding pin can be used as input. Double click on 'IN' box; the caption will change to 'OUT.' Now, the pin can be

used as output pin.

Controlling devices

Make all the eight pins as OUT, then connect a few LEDs on the corresponding pins on the Raspberry Pi board. Click on each pin, say pin 13 (GPIO 27) to begin with, on the browser screen; it will change from black to orange and the corresponding LED will glow.